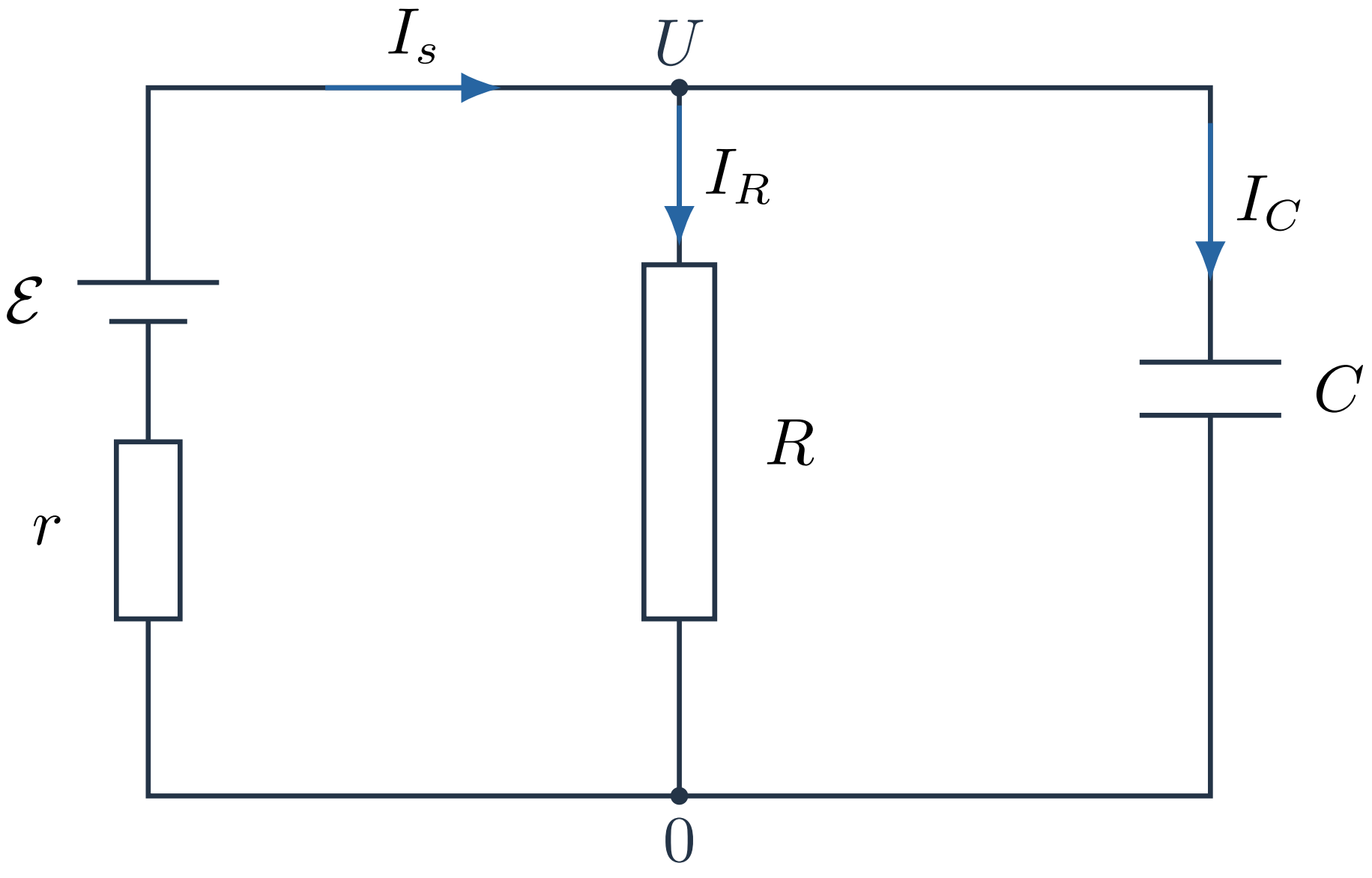

$$I_s = I_R + C\,dU/dt$$